

# Munawar Kazmi

Robotics and AI Engineer

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## Profile

Robotics and AI engineer (MSc AI and Robotics, Commendation). I build ROS2 planners, embedded firmware and verification layers for robots that plan with language models, and I have two systems running in the field. Every number below is reproducible from a public, CI-tested repository. Open to roles in the UK, EU and USA.

## Technical Skills

**Robotics and systems:** C++20, Python, ROS2 (Nav2, plugin development), differential and property-based testing, CI/CD, Docker

**AI and simulation:** PyTorch, TensorFlow, TensorRT, LLM evaluation and integration, Isaac Sim, MuJoCo, Gazebo, Unity

**Embedded and full-stack:** ESP32, LoRa mesh, bare-metal C/C++, Jetson Nano, React, TypeScript, PostgreSQL

## Experience

**Full-Stack Software Engineer (Contract), Safina International** Multan, Pakistan; Jul 2026 - present

- Sole engineer of the institute's live management platform: **seven user roles**, prorated billing, payroll and an append-only audit trail across **33 tables** under PostgreSQL row-level security.

**Embedded Systems Engineer (Contract), Muxtronics** Multan, Pakistan; Jan 2025 - May 2025

- Wrote deterministic C++ firmware for real-time image processing on ESP32-CAM for a commercial intrusion-detection prototype: 800x600 frames reduced to a **16x12 decision grid**, no ML and no vision libraries.
- Designed the LoRa mesh protocol that carried alerts off the device.

**Robotics Integration Specialist, Robot House, Univ. of Hertfordshire** UK; Jan 2024 - May 2024

- Built the gesture recognition and motion planning pipeline for the JD humanoid in Unity and C++, integrated through ROS2, and deployed PyTorch models onto its real-time control loop for a 20-participant user study.

**Research Associate, University of Hertfordshire** Hatfield, UK; May 2023 - Dec 2023

- Optimised TensorFlow models for real-time inference on the Unitree Go1 quadruped and integrated them into its control loop, within the latency budget a walking controller leaves for perception.

**Visiting Lecturer, UCLTILS and Pak-Turk Maarif Schools** Multan, Pakistan; Aug 2025 - present  
Computing Principles and Logic on the University of London BSc Artificial Intelligence (from Sep 2026), and an applied ML curriculum for 30+ school students.

**Embedded Systems Intern, National Centre of Robotics and Automation** Islamabad; Mar 2022 - Apr 2022  
Optimised bare-metal C firmware for embedded controllers, validated across 100+ safety-critical test cycles.

## Engineering Projects

**ROS2 path planning: A\* and D\* Lite for Nav2** github.com/munawarkazmi/ros2-dynamic-path-planning

- ROS-free C++20 planning core with Nav2 plugin adapters. Over a seeded 200-trial blocked-route benchmark D\* Lite replans **4.3x faster on average**, with optimality fuzz-validated against Dijkstra over **185,237 replans**.

**LLM safety verifier and llm-nav-shield** github.com/munawarkazmi/llm-nav-shield

- A deterministic gate between LLM planners and Nav2 caught **35/35 and 32/32 unsafe proposals** from two models with zero misses; a provably correct planner then recovered **38/38** to verified-safe paths and halted in **10/10** cases where no safe path exists. Every number is replayed in CI.

**exact-predicates and toolcall-contract** github.com/munawarkazmi/exact-predicates

- Exact geometric predicates in C++, grown from a real D\* Lite bug: **657 adversarial cases** where floating point is wrong and the exact version is right. A contract checker for LLM tool calls: of 40 model calls a schema check passes 39 and the contract passes 32.

## Research

**legibility-bounds** (sole author, submitted to IEEE RA-L, 2026): certified upper bounds on legible robot motion; 32 of 32 intervals hold with zero violations. DOI 10.5281/zenodo.21834955.

**plan-failure-bench** (sole author, working paper, 2026): 60 proof-carrying instructions and 548 CI tests measuring how LLM planners fail, with no human or LLM judge. DOI 10.5281/zenodo.21756817.

**MSc thesis** (2025): speech against gesture on the JD humanoid,  $N = 20$ ; a significant clarity effect,  $d = 0.58$ ,  $p = 0.017$ .

## Education

**MSc Artificial Intelligence and Robotics, Commendation, University of Hertfordshire** 2023 - 2024

**BEng Mechatronics Engineering, National University of Sciences and Technology (NUST)** 2018 - 2022